

The observer in a modified version of the multiple worlds interpretation

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ABSTRACT. In [1], a derivation of the Born rule was presented. In this paper, the definition and role of the observer is clarified.

RÉSUMÉ.

P.A.C.S.: 04.60.-m; 03.65.Ta

1 Introduction

In [1], a derivation of the Born rule was presented. This derivation was performed in the framework of the multiple worlds interpretation (MWI). This framework followed the line of thinking as originally set forth by Everett in his relative state formulation, with the point of departure being to replace the Born rule with outcome counting. It was then demonstrated that if one assumes a certain set of postulates to apply in the low energy limit, then the Born rule is recovered as a coarse grained approximation.

The purpose of this paper is to clarify the notion of the observer in the model outlined in [1]. There are three criteria that the definition of the observer should meet. The first criterion is that it should be compatible with the essential ontology of the MWI, according to which *any physical object* can play the role of the observer. That is to say that the “observer,” according to the essential ontology of the MWI, could be a human being, a (non-conscious) experimental apparatus, or even a subatomic particle. Recall that Everett initially divided the universe into two subsystems, the observer and the observed system. The observer is

allowed to be chosen arbitrarily as any physical object, and the wavefunction of the observed system is then defined *relative to* the observer (hence then name “relative state” formulation).

The second criterion is that the notion of outcome counting must be applied to the observer, not to the system under observation. In the words of Everett’s original formulation, an observer in a definite initial state evolves into a final state represented “in terms of a superposition, each element of which contains a definite *observer state*” [2] (emphasis mine). For this to make any sense, there must be instances in which the time-dependent evolution of the observer provides a *multiplicity* of solutions, each of which is physically distinct.

The third criterion is that our definition of the observer should be done in a manner that tells us how to define the ensemble of spin foams discussed in [1].

2 Observer state space

“The state of one subsystem [i.e., the system under observation] does not have an independent existence, but is *fixed* only by the state of the remaining subsystem [i.e., the observer]” [2] (emphasis mine). How might this be achieved mathematically? Let us propose how this might be done.

Recall that a quantum mechanical world is represented mathematically as a spin foam. Let us assume that there exists a continuum limit to these spin foams, such that as the distance between nodes approaches zero, the spin foam becomes a smooth manifold. A spin foam in which nodes are separated by some small average distance ϵ , therefore, can be viewed as an approximation, with the level of approximation improving as ϵ approaches zero. We may next ask: what is the set of “all possible” manifolds allowed by the theory?

Let us entertain the notion that the metric must be everywhere analytic and nowhere singular. Pick an individual manifold, and consider next the metric g_{ij} and its power series at an individual point p ¹. By the method of analytic continuation, it is possible to calculate an *exact solution* over the entire manifold, given only the metric and its power series at the point p ². For this to be unique, however, the topology of

¹We must assume the axiom of choice to make these assignments.

²We make the assumption that the domain of analyticity must extend over the entire manifold.

the manifold must be prespecified. In other words, if we are given points p_1 in manifold M_1 and p_2 in M_2 , where the manifolds have the same topology, then if we assume that the metric and power series is the same for each point, then $M_1 = M_2$.

This suggests the following notion. First, define the “observer” as a node on the graph. As the foam becomes a manifold in the continuum limit, then the observer goes from a node to a point p ; or more specifically, with the metric and power series at that point. The ensemble associated with p is then defined as the set of all manifolds that are possible solutions to the global metric. From the above, we may conclude that any two manifolds in E that are of the same topology must be two copies of the same manifold; and if the two manifolds are distinct, then they must be of distinct topology. Therefore the set of all possible ensembles associated with an observer p could be restated as the set of all possible topologies. We note that if a given point p appears n times in a manifold, then that manifold appears n times in the ensemble.

Definition 1 *Q is the set of “all possible” manifolds that are admitted by the theory. It could also be called the set of “all possible topologies.”*

Although there are only sixteen components to g_{ij} ³, there are of course an infinite number of components involved in providing its power series. It follows that observer state space is infinite dimensional.

We note in passing that the ensemble E corresponding to a particular point p in state space is determined entirely by p ; in particular, there is *not* any additional dependence on the past history of the observer.

3 Time dependent evolution

Suppose that we are approximating state space by only considering, say, the first d decimal places (with d some suitably very large integer) of each component of each state in state space. Given any point p in state space, we can draw a collection of directed links, each of which starts at p and ends at some other point in state space. In this manner, a state machine can be constructed. A series of links joined head to tail in series

³If we assume that our manifolds are governed by general relativity – which we have *not* yet done! – we may further assume that the metric is symmetric, so that there are only ten components required to specify g_{ij} . It is interesting to speculate that these ten components correspond in some way to the ten dimensions of string theory.

is referred to as a path. Starting at any given point p , the set of all paths that start at p will construct a branching tree-like diagram, as in the Figure. We will call this the *transition matrix*. This branching tree is meant to correspond exactly with the branching tree like diagrams that are often discussed in the context of the standard MWI.⁴ According to the model proposed in [1], the probability that an observer starting at state p will evolve along a particular branch is determined by the principle of outcome counting. It is noted in passing that the time dependent evolution is determined entirely by the *local* structure of the tree diagram within an infinitesimal ball around the point p in state space.

Note that the probability that the state p_1 will evolve into state p_2 is determined entirely by the makeup of the transition matrix⁵. Indeed, starting at point p , the relative probability of evolution to point p_i is proportional to the number of distinct paths linking p to p_i .

In the limit as d approaches infinity, state space becomes continuous, and each path becomes a geodesic. In other words, the evolution of the observer through (proper) time is calculated by the parallel-transport law, i.e. according to the geodesic equations[3]:

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0 \quad (1)$$

We stipulate that there are no “dead ends,” ie there are no points p in state space that do not have any possible future evolution.

Consider the fact that, according to standard QM, the probability of evolution from one state to another state is *never strictly zero*. This suggests that we require that, given two states p_1 and p_2 , there is always at least one path from p_1 to p_2 . Since the reverse is also true, then given any state p , there is always at least one path that starts and ends at p .

Consider two paths p_1 and p_2 such that p_2 is downstream of p_1 .

Theorem 1 $E_{p_2} \subset E_{p_1}$.

Corollary 1 For any p , $E_p \subset E_p$.

⁴Just like in the MWI, merging is allowed as well as branching.

⁵The state diagram and transition matrix concept may be compared to Wolfram’s cellular automata (CA).

In other words, it's fractal. Therefore the number of copies of any given manifold is infinite. So when we ask: how many copies of M are in the ensemble E ? the answer is: infinite number of copies. The information content comes in asking: what is the *ratio* of the number of M_1 to M_2 in E ? This is really what distinguishes one ensemble from another.

And probability can be calculated in two ways: the outcome counting way; also, the count the measure of manifolds way. The thing is that the measure of manifolds is in the eye of the beholder, so to speak. I think.

4 Hilbert space

We will define addition, multiplication, and an inner product over observer state space and demonstrate that these define a Hilbert space.

The elements of Hilbert space $\mathbb{P}$ are the points p . For any p , there corresponds an ensemble E_p ; so E could alternatively be taken to be an element of Hilbert space. That is to say that we have two Hilbert space representations: the *ensemble representation* $\mathbb{E}$ and the *point representation* $\mathbb{P}$. In the ensemble representation, the elements E are defined as the *power set* of Q .

In $\mathbb{E}$, we can define the empty ensemble $\emptyset$ as well as negative ensembles that have a “negative” number of manifolds in them. There are no counterparts to these in $\mathbb{P}$.

Remark 1 *For any given p , there corresponds a unique E : $\mathbb{P} \subset \mathbb{E}$.*

This is pretty obvious.

Conjecture 1 *Given any E , it is possible to expand E as a sum of points $\sum p_j$.*

Heuristic proof. Any given manifold is just a collection of points p .

Definition 2 *Addition $+$. $p_1 + p_2 = E_{p_1} \cup E_{p_2}$, where $\cup$ is the usual notion of the union of sets. Note that there can be multiple copies of any given individual manifold within an ensemble.*

Commutation under addition follows from the definition of union.

Definition 3 *Multiplication $*$. $k * p = \cup_k E_p$. That is, k copies of E .*

Since there are an infinite number of manifolds in any E , then we can eg multiply by one-half to obtain half the number of each manifold. Which produces the same ensemble, but at a different scale. If k is negative, think of this as negative number of copies of each manifold in the ensemble; thus, $p - p = E + (-E) = \emptyset$. It is obvious that this is closed under multiplication because no matter how you manipulate the ensemble, you always just end up with another ensemble of manifolds.

Definition 4 *Inner product $\langle p, p_i \rangle$. An inner product is always defined with respect to a given basis, which is any (finite or infinite) set of points p_i . The projection of p onto p_i is $\langle p, p_i \rangle$ and is defined as the relative probability that the observer starting at p will end up at p_i before arriving at any other basis $p_j, i \neq j$.*

Probability is defined by outcome counting applied to the transition matrix as discussed above. By definition, $\langle p, p \rangle = 1$ since any observer will end up “at itself” before any other point, since it’s already there. So for any set of basis points, $\langle p_i, p_j \rangle = 1$ for $i = j$ and 0 for $i \neq j$. Thus, the basis points are automatically normalized and orthogonal. It also follows that any wavefunction p is automatically normalized.

Any wavefunction can be expanded in terms of any basis: $p = \sum |c_i|^2 p_i = \cup |c_i|^2 E_i$ where $|c_i|^2$ is the probability of collapse onto p_i , that is, $|c_i|^2 = \langle p, p_i \rangle$. (In the low energy approximation, we show that $c_i = \chi_{s,d_i}^1$.)

Actuallyk it’s possible for a basis to be incomplete, if the probability that it never hits one of the basis points is nonzero. Is this possible in a finite state space? Not sure.

An operator A inputs a basis and outputs the value of an observable that is true for that basis. For example, if p_3 represents the state that the particle was seen by detector 3, then we would have $Ap_3 = 3$. The basis points are eigenfunctions, and the observable is the eigenvalue. In general, $Ap_i = a_i$. (Compare this to standard QM def of an operator, where $Ap_i = a_i p_i$.) An operator is always defined with respect to a particular basis!

We assume that the operator is linear, ie, $A[ap_1 + bp_2] = aAp_1 + bAp_2$ where a, b are real numbers. So we have: $Ap = A[\sum |c_i|^2 p_i] = \sum |c_i|^2 Ap_i = \sum |c_i|^2 a_i$ which is, by definition, the expectation value of that observable, which we call $\langle A \rangle$.

Thus, $\langle A \rangle = Ap$. (Compare this to standard QM, where $\langle A \rangle = \langle p, Ap \rangle$.)

If we want to define the observer as “my brain in some definite state,” then we just take the ensemble $E = \sum p_j$ which is the sum of each individual point in my brain. We can calculate the expectation value $\langle A \rangle = AE = A[\sum p_j] = \sum Ap_j$ which is well-defined! So we come full circle: *any* physical object can play the role of “observer.”

It is trivial to show that the inner product is symmetric. $\langle p, p_i \rangle = |c_i|^2$ We also have $\langle p_i, p \rangle = \langle p_i, \sum |c_j|^2 p_j \rangle = \sum |c_j|^2 \langle p_i, p_j \rangle = |c_i|^2$ Thus, we have $\langle p, p_i \rangle = \langle p_i, p \rangle$

5 Low energy approximation

What does “low energy” mean in this model?

5.1 definitions of the low energy approximation

One: an area of state space where there is branching more than merging. (second law of thermodynamics).

Two: places in state space where the lower-derivative components of the metric are close to zero. That is, smaller than the higher order components, since there’s no such thing as “almost zero” when you haven’t defined units! That means that the large-scale structure of manifolds is flat, whereas the small-scale structure is warped and twisted, ie, microscopic wormholes. Of course, the approximation term d places an upper bound on the frequency of the warpage/twistage of spacetime, which translates into a lower bound on the size of wormholes. As d becomes infinite, we get progressively smaller and smaller microscopic wormholes!

Three: all of the stuff discussed in [1].

5.2 strategies to demonstrate low energy approximation

We should show that states tend to gravitate towards low energy states; low energy states are “attractors.” To show this, we show that given any p , and given a p_h and p_l , where p_h and p_l meet the requirements of “high” and “low” energy as defined using (any of) the above definitions, that:

$$\langle p, p_l \rangle \gg \langle p, p_h \rangle \quad (2)$$

where the basis states are defined as: p_h and p_l .

The inner products can be calculated using one of three methods: ration of measures of the ensembles; count the number of analytic curves from p to p_l (or p_l); serial multiplication of terms as you go down a path.

Another argument. Consider the set of all possible analytic curves. Given d as defined above (the number of derivatives that we consider; like a sampling rate), and a similar parameter r where $d \gg r \gg 0$. Both of these effectively serve as frequency cutoffs. The measure of the ones where all frequencies less than r go to zero at infinity proper time, and where frequencies higher than r get maxed out (max freq is one half of d , the sampling rate!) is greater than the measure of the ones where they are all big magnitude, because of an argument similar to Feynman's for all possible paths. It's the origin of a minimization law! I should be able to do a numerical calculation of this. O THAT would be cool ...

6 more musings

Slight redefinition of inner product. Define $\langle s_i, s_j \rangle$ as the probability that, if you start at s_i , then you will end at s_j *before* coming back around to s_i . This definition does NOT depend on the choice of a "basis."

So now let's reconceptualize basis. A set of B states $s^b, b \in [1, B]$ forms a basis over the domain D if and only if the following hold for any $s_i \in D$:

Complete: $\sum_b \langle s_i, s^b \rangle \sim 1$

Orthogonal: Two ways to state. One is that the probability of starting at s_i and coming back to s_i BEFORE hitting any of the s^b is (approximately?) zero. Another is to say that the probability of going from s_i to one s^b to another s^b (then back to s_i ?) is (approximately?) zero.

If the "approximately" conditions turn out to be necessary, then perhaps another way of stating the low energy approximation is to state that the state s_i is low energy iff "there exists at least one good basis over the domain s_i ."

If we require all bases to conform to the above more stringent requirement, then the inner product as initially defined is equal to the inner product as currently redefined.

From this, we see that $\langle s_i, s_j \rangle$ is basis independent. So we can say that $\langle s_i, s_j \rangle + \langle s_j, s_i \rangle \in (0, 2]$ is a conserved quantity, independent of basis, and we can use this to define the distance between any two states.

So now the question becomes: how to define the norm?

Acknowledgments:

Thanks everyone!

References

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(Manuscrit reçu le date)